DAA – MINIMUM MERGED QUESTION BANK

Design and Analysis of Algorithms (410241) | Based on 4 uploaded In-Sem papers

Counting rule: Every original question/sub-question occurrence is assigned to exactly ONE merged entry. Therefore, there is no overlapping or double counting. Similar questions are merged aggressively, but questions requiring a materially different proof, algorithm, or concept remain separate.

1. Iterative Algorithms and Design Issues

What is an iterative algorithm? Explain iterative algorithm design issues using suitable examples. Also analyze iterative-design issues in a given iterative algorithm such as linear search.

Exact merged frequency: 5 source question/sub-question occurrences

2. Algorithm Correctness Using Loop Invariants

Explain why algorithm correctness is important. Define loop invariant and use it to prove the correctness of a given algorithm, such as finding the summation of n numbers.

Exact merged frequency: 2 source question/sub-question occurrences

3. Methods of Proving Algorithm Correctness

Explain how to prove an algorithm correct, including the use of counterexamples. Prove the correctness of a given recursive or iterative algorithm, such as minimum element, factorial, or square of a number, using a suitable method.

Exact merged frequency: 5 source question/sub-question occurrences

4. Problem-Solving Strategies

Explain / write a short note on any four problem-solving strategies.

Exact merged frequency: 3 source question/sub-question occurrences

5. Importance of Studying Algorithms and Their Relation to Technology

Why do we need to study algorithms even with very fast computers and hypothetically unlimited memory? How are algorithms related to technology?

Exact merged frequency: 2 source question/sub-question occurrences

6. Algorithm Efficiency and Case Analysis

Explain and apply best-case, average-case and worst-case analysis. Analyze the efficiency/complexity of a given algorithm such as insertion sort, linear search, or selection sort. Discuss limitations of best-case analysis and whether average-case efficiency is the average of best and worst cases.

Exact merged frequency: 6 source question/sub-question occurrences

7. Asymptotic Notation and Its Properties

Explain and apply O, Ω and Θ notation. Discuss properties and implications such as whether $f(n)=O(g(n))$ implies $g(n)=O(f(n))$, what simultaneous $O(f(n))$ and $\Omega(g(n))$ bounds imply, and justify whether a given expression belongs to a stated asymptotic class.

Exact merged frequency: 4 source question/sub-question occurrences

8. P, NP, NP-Hard and NP-Complete Problems

Explain P, NP, NP-Hard and NP-Complete problems/classes in computational complexity theory with suitable examples.

Exact merged frequency: 4 source question/sub-question occurrences

9. Polynomial-Time Reducibility

What is polynomial-time reducibility? Explain its importance in computational complexity theory.

Exact merged frequency: 1 source question/sub-question occurrence

10. Vertex Cover as an NP-Complete Problem

What is an NP-Complete problem? Explain/prove that the Vertex Cover problem is NP-Complete.

Exact merged frequency: 2 source question/sub-question occurrences

11. SAT and 3-SAT

Explain SAT and 3-SAT using suitable examples. Discuss why SAT is important in theoretical computer science and prove/discuss the NP-Completeness of 3-SAT.

Exact merged frequency: 3 source question/sub-question occurrences

FREQUENCY SUMMARY – NO OVERLAP

Each original occurrence is counted once only. The total below is 37 source question/sub-question occurrences.

Rank	Merged question / concept	Exact frequency
1	Algorithm Efficiency and Case Analysis	6
2	Iterative Algorithms and Design Issues	5
3	Methods of Proving Algorithm Correctness	5
4	Asymptotic Notation and Its Properties	4
5	P, NP, NP-Hard and NP-Complete Problems	4
6	Problem-Solving Strategies	3
7	1SAT and 3-SAT	3
8	Algorithm Correctness Using Loop Invariants	2
9	Importance of Studying Algorithms and Their Relation to Technology	2
10	Vertex Cover as an NP-Complete Problem	2
11	Polynomial-Time Reducibility	1

Source mapping check: The 4 papers contain 37 counted question/sub-question occurrences under this extraction method, and the corrected merged frequencies sum to 37: 5 + 2 + 5 + 3 + 2 + 6 + 4 + 4 + 1 + 2 + 3 = 37.